MyBreakPoint Match Coach: Bridging the Coaching Gap with an LLM-Driven Intelligent Tutoring System for Tennis Strategy

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Abstract

While technology has revolutionized the tracking of physical and statistical performance in sports, tactical and strategic decision-making remains difficult for amateur athletes to train independently. Tennis requires high levels of metacognition under immense time pressure and cognitive load. MyBreakPoint Match Coach is an Intelligent Tutoring System (ITS) designed to bridge the “coaching gap” by providing real-time, scaffolded strategic feedback during a match. By combining a structured tennis Domain Model with a Large Language Model (LLM) acting as the Pedagogical Model, the system evaluates raw match data to track student mastery of specific Knowledge Components (KCs). Unlike traditional template-based applications, this architecture successfully handles the “composition problem” of strategy, transforming passive statistics into active, just-in-time coaching.

Problem

Tennis is a physically demanding sport, but matches frequently turn on mental decisions made under extreme pressure. Players must manage high cognitive load between points, reflecting on what just happened and adjusting their tactics accordingly. This requires highly developed metacognition and tactical reasoning.

Unfortunately, there is a pervasive “coaching gap” in amateur and junior tennis. While professional players have dedicated coaches to help manage this cognitive load, most junior and club players walk onto the court alone. Without a coach, players are left with raw statistics (e.g., first serve percentages, unforced error counts) but lack the pedagogical guidance to connect those numbers to actionable strategy.

A technology-based ITS is highly suitable for this problem because spectators or players can easily log basic match events. However, raw data is insufficient for learning. Players need an “Inner Loop” system capable of interpreting messy data in real-time, identifying strategic errors, and providing short, specific feedback before the next point begins.

Background

The current landscape of sports technology is heavily focused on data collection rather than pedagogical instruction. Systems like SwingVision or Sloun provide stat-heavy analytics and heat maps. However, in the context of tutoring architectures, these applications only fulfill “Outer Loop” functions (VanLehn, 2006). They track the state of the game but lack the “Inner Loop” support necessary to guide a player’s strategic adjustments step-by-step. Similarly, video analysis tools like Dartfish are excellent for cognitive skill acquisition regarding physical mechanics, but they do not address real-time tactical reasoning.

Transitioning from data to dialogue requires an intelligent agent. Historically, dialogue-based ITSs and rule-based cognitive tutors have required massive engineering efforts to build (Aleven, 2010). Recently, Large Language Models (LLMs) have shown promise in generating education hints rapidly (Pardos & Bhandari, 2024). However, general LLMs lack a grounded Domain Model. Without structured, domain-specific constraints, they are prone to providing generic advice rather than the specific, formative feedback required to change a learner’s behavior (Shute, 2008). Furthermore, simple template-based hint generators struggle with the “composition problem”, the challenge of assembling isolated plans and events into a cohesive strategy (Lane, 2005). MyBreakPoint Match Coach builds upon these foundations by using structured Knowledge Components to ground an LLM, creating an adaptive pedagogical agent for athletic strategy.

Approach

The primary learning goal of MyBreakPoint Match Coach is to improve a player’s metacognitive awareness and strategic reasoning during a match. The desired outcome is for the player to internalize tactical patterns so they can eventually self-correct without the tutor’s intervention.

To achieve this, the system leverages a hybrid AI architecture:

1. **The Domain Model**: A structured library of tennis strategy rules representing core Knowledge Components (KCs), such as “Second Serve Placement) or “Return Aggression." This provides the ground truth that a general LLM lacks.
2. **The Pedagogical Model**: The system utilizes an LLM (running via Cloudflare Workers AI) as the “coach.” The LLM evaluates raw, compressed match data against the Domain Model to decide when a strategic error is significant enough to trigger a just-in-time hint.
3. **Learner Modeling**: The system tracks success rates on specific KCs. If a player repeatedly fails to execute a strategy (e.g., continually hitting to an opponent’s forehand), the system detects a lack of mastery. It then utilizes scaffolding, increasing the specificity of its feedback from facilitative hints to more directive, actionable commands until the player adjusts.

Results: Description of System

MyBreakPoint Match Coach is designed to be seamless for the learner, prioritizing low-effort logging to keep the user’s focus on the game.

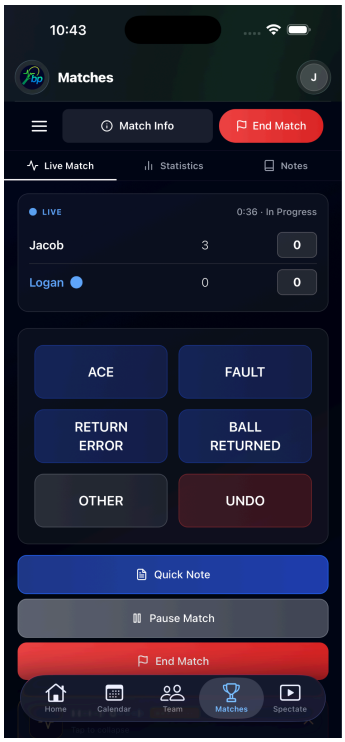

Figure 1: The logging interface, designed for rapid input during the 20-second window between points.

During a match, as points are logged, the system compresses the data into a focused summary to prevent overwhelming the LLM with a massive JSON dump. When a predefined milestone is reached (e.g., changeovers, a set break, or a specific error threshold), the Match Coach is triggered.

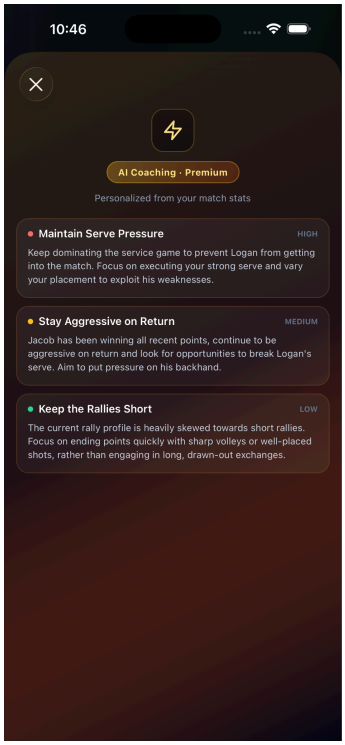

Figure 2: Scaffolded live feedback presenting directive hints based on the player's immediate KC gaps.

To ensure reliability and prevent the UI from breaking, the LLM operates under strict structural prompts. The output is run through automatic rule checks (validating JSON structure, lengths, and required fields before it is cached in Cloudflare D1. This validation ensures the pipeline produces usable tutor output, while the underlying Domain Model ensures the advice remains strategically sound and relevant to the specific match context. After the match, the system generates a structured training plan detailing what went well and providing concrete drills to support continued reflection.

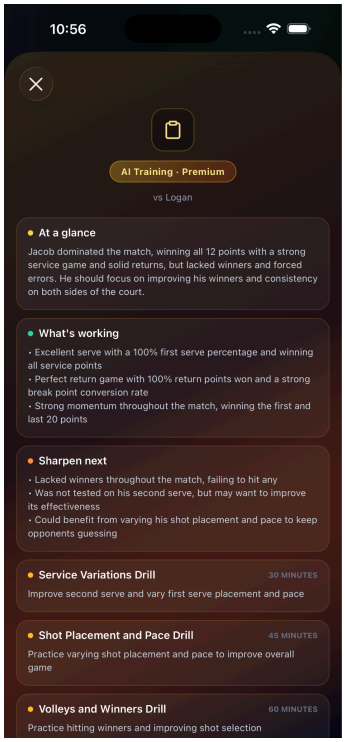

Figure 3: The post-match training plan, which supports long-term skill acquisition by providing structured reflection and concrete drills targeted at the specific Knowledge Component (KC) gaps identified during gameplay.

Next Steps

If I were to continue developing MyBreakPoint Match Coach, the immediate next step would be a formal evaluation study. I would conduct an A/B test with junior players comparing the efficacy of the ITS against traditional stat-taking apps (like SwingVision). The research questions would focus on whether MyBreakPoint Match Coach reduces cognitive load and whether players demonstrate faster tactical adjustments (shorter learning curves for strategic KCs) when using the system.

A current weakness of the system is its reliance on manual data logging, which can result in incomplete or “messy” data if the spectator misses a point. To extend the project, I would explore integrating computer vision (processing video feeds to auto-log points) or physiological monitoring (using smartwatch heart-rate data to gauge player arousal and frustration). This would allow the Pedagogical Model to factor in the player’s affective state, providing empathetic support when a player is in a state of cognitive disequilibrium.

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